

10.6 APPARATUSES USED FOR EXTRACTION OF METALLIC COMPONENTS

Three groups of experts were working on the extraction of metals from the ore. First are the metalworking engineers and metalsmiths. They mine the ore and extract the metal. The second group was the Āyurvedic practitioners. They also extracted metals and prepared therapeutic formulations using the powder form of the metals (bhasma). Thirdly the alchemists were involved in extracting metals and used them in their alchemical preparations. While the second and third groups might have worked on a smaller scale, the metalworking engineers engaged on a bigger scale. Nevertheless, the process and requirements were similar, and they have been adopting similar processes for extraction.

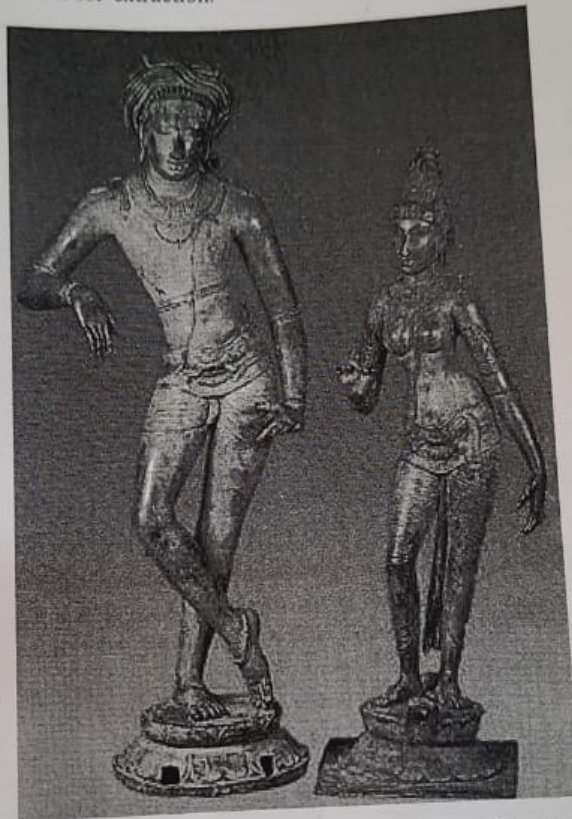


FIGURE 10.9 A 11th Century CE Bronze Idol of Śiva-Pārvatī
Source: https://upload.wikimedia.org/wikipedia/commons/9/94/Bronze_siva.png

A closer look at the alchemy works belonging to the early centuries of the Common Era gives us a good indication how a laboratory of a scientist looked like and what type of apparatuses were fabricated by them and used for extraction of various chemicals and metallic components. Several apparatus and appliances called yantras were employed by the alchemists.

For instance, RRS has described a variety of crucibles (Mūṣā) and furnaces for the manufacture of Āyurvedic medicines. It also describes in total 51 kinds of metallurgical tools (upakaraṇas), 36 kinds of equipments (yantras), 17 types of crucibles, and 9 types of furnaces. The alchemist, the Āyurvedic expert, and the metalworking engineer seem to be benefiting from each other's know-how and used similar apparatuses, albeit in varying sizes to suit their requirements. We have already seen some of these in the previous sections. However, we shall discuss this aspect in a focused manner here.

The crucible, known as mūṣā-yantra is the frequently used apparatus by the alchemists as it serves as a generic container for a variety of operations. The mūṣā-yantra was predominantly earthen. To prepare the crucibles, the earth of anthill, rice husk, iron rust, chalk, human hair, and a few other ingredients are mixed in a certain proportion, and to this goat's milk is added. The mixture is kneaded into a dough-like mass which would then be shaped into the desired forms, and finally sun-dried³³. There were various types of crucibles for different operations. For the extraction of zinc from calamine, a crucible of the shape of brinjal (Solanum melongena) is used. To this crucible, a tubular end is added, which expands towards its mouth like a flower.

We shall see in brief some of the apparatuses found in the ancient Indian laboratory³⁴. The apparatus is simple and made of earthen material shaped into different forms. In all the cases the source of heat was either cow-dung or wood. As we know, this may not yield a high temperature, and therefore wherever required the substances are heated for a few days to in some cases weeks.

Dolā-yantra uses a suspension mechanism using which the ingredient to be subjected to some treatment is kept in a piece of cloth, tied, and suspended using a rod. This is immersed in a pot half-filled with the desired liquid. The liquid is then heated from the outside. See Figure 10.10(a) for an illustrative sketch of Dolā-yantra.



FIGURE 10.10 Examples of Yantras Used by Alchemists in India

Source: Sen, S.N. and Subbarayappa, B.V. (2009). *A Concise History of Science in India*, Indian National Science Academy, 2nd ed., pp. 426–430. Reproduced with permission.

Svedani-yantra is used for steaming purposes. The mouth of a pot is covered with a piece of cloth and the substance to be steamed is placed on the cloth. Water is kept in the pot and is covered using another pot. The substance is steamed by boiling the water.

Pātana-yantra is used whenever a sublimation or distillation process is involved. It consists of a well-baked pot of suitable dimensions. Water is filled in this pot to its neck and another bigger pot is kept over it in an inverted position. The two necks of the pots are firmly sealed using an adamantine made of lime, raw sugar, and buffalo's milk. A downward sublimation process is applied as follows. When the substance is smeared inside the upper vessel and heated from outside by burning cow-dung cakes, it vaporizes and collects in the water in the lower pot. On the other hand, the substance may also be heated in the lower pot and the upper one is kept cooled by applying wet rags. The sublimate deposits in the interior of the upper pot. Figure 10.10(b) is a schematic representation of a version of Pātana-yantra for the downward distillation process. In this version, the vapour passes through a pipe into another pot kept alongside. The second pot is cooled with water and the vapour condenses into liquid and is collected in the second pot.

Dhekī-yantra is also used for the distillation of substances such as mercury. A hole is made slightly below the neck of a pot and one end of a bamboo tube is introduced into it. The other end is fitted into a brass vessel that is filled with water. Mercury mixed with the other desired substances is subjected to distillation in this type of apparatus.

Vālukā-yantra is of a type of sand-bath for heating substances uniformly and for usually a long time. A long-necked bottle, containing the substance to be heated, is kept buried in sand in an earthen pot up to three-fourth of its height. The apparatus is heated from below for a long time till a straw when placed on the surface of the sand, catches fire. Instead of sand, sometimes salt is used.

Dhūpa-yantra is used for fumigation purposes. Fumigation of the leaves of gold or silver with the fumes of sulphur or arsenic substances is done in this apparatus. Two vessels are employed for this purpose. In the lower vessel, iron bars are placed in a slanting position below its mouth, and leaves of gold or silver are placed on them. Sulphur or the arsenic substance is placed in the lower vessel and the other vessel is used for covering purposes. On heating, the fumigation takes place.

For heating mercury and sulphur together, the following details of an mūṣā-yantra are given in the Rasārṇava³⁵: The apparatus consists of two crucibles; one of them has a narrow orifice. In one crucible sulphur is taken and in the other mercury. Mercury and sulphur are to be moistened with filtered garlic juice. The crucible containing sulphur is to be inserted into the one having mercury, and the apparatus carefully lowered into an earthen pot over which another earthen pot is to be kept, and the rims luted with a cloth. It is heated from outside by cow-dung fire for three days.

- ♦ Yuktikalpataru an 11th century CE work mentions the relative properties of iron-carbon alloys produced in different regions of India.
- ♦ The alchemist, the Āyurvedic expert, and the metalworking engineer seem to be benefiting from each other's know-how.

SUMMARY

- ▶ Indian civilisation has a rich heritage spanning over the recorded history of the last two millennia. Moreover, there are several other archaeological and literary resources that point to the existence of a vibrant community in antiquity.
- ▶ In technology, ancient Indian's contributions to metalworking, and in more general terms to metallurgy and material science are noteworthy.
- ▶ The high status of iron and steel technology in ancient and medieval India is reflected in the manufacture and use of numerous large iron objects, including forge-welded cannons.
- ▶ A long and flourishing tradition of the copper industry in ancient India can be inferred from a number of archaeological specimens.
- ▶ Extensive archaeological excavations conducted in several parts of Rajasthan clearly show that the earliest artifact containing an appreciable amount of zinc anywhere in the world is from India.
- ▶ The Vedic corpus has multiple references to iron, copper, gold, silver, and tin. There are a few important treatises pertaining to the 5th century CE to 13th century CE that deal with metallurgy and metalworking.
- ▶ Ancient Indians developed skills to explore ore deposits and mine portions that are remunerative.
- ▶ Rasa-ratna-samuccaya has mentioned many processes for obtaining pure copper from chalcopyrite by smelting small charges in crucibles. In Āyurveda, copper in the powder form (bhasma) is used for therapeutic formulations.
- ▶ Ancient craftsmen could subject the purified gold metal bar to metalworking techniques such as forging, punching, embossing, etc. so that ornaments of intricate designs could be produced.
- ▶ Studies on the Zawar zinc mines provide rich insights into zinc mining in ancient India. Ancient Indians adopted a novel technique of downward drift reduction distillation process for zinc production, which is a precursor to the modern processes adopted all over the world.
- ▶ The compound HgS is referred to as Rasa-sindūra in the Āyurvedic text Caraka-saṁhitā. Hg, as well as HgS, have been extensively used in the preparation of Āyurvedic medicines.
- ▶ Ancient Indians have been extracting iron from ore deposits and producing sponge iron or wrought iron using a variety of furnaces.
- ▶ The carburisation of hot sponge or bloom could be done by selective carburisation of cutting edge of the implements by the application of carburising paste and reheating to 950°C to 1000°C. This method was followed by ancient Indians to develop sharp cutting edges such as surgical knives and swords.
- ▶ A closer look at the alchemy works belonging to the early centuries of the Common Era gives us a good indication of how a laboratory of a scientist looked like and what type of apparatuses were fabricated by them and used for extraction of various chemicals and metallic components.
- ▶ The beautiful idols and icons made during the Chola era were produced using bronze using the lost wax casting technique.

REVIEW QUESTIONS

1. What are the ways by which one can make an assessment of the S & T heritage of an ancient culture?
2. Do you think that ancient Indians had a good understanding of mining and metallurgy? Is there archaeological evidence to support this?
3. How can we infer ancient Indian's knowledge and use of iron and copper?
4. What is the status of zinc with respect to ancient Indian society?